

BRAF fusion-hotspot benchmark report

Study: msk_impact_2017

Abstract

This report analyzes BRAF gene fusions from the msk_impact_2017 study, covering 41 fusion events. 33/41 fusions (80.5%) were in-frame. 33/41 fusions (80.5%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test ($p=0.00060718$, statistically significant at $\alpha=0.05$). Compared to the literature benchmark (PMC5461196: 100.0% in-frame, 100.0% domain-retained), this run observed 80.5% in-frame and 80.5% domain retention.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 41 analyzed fusion events, identifying 25 distinct fusion partner genes. The most frequent partner was SND1, observed in 8 events.

Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 31/33 (93.9%) of in-frame fusions retained the domain, $p=0.00060718$ (statistically significant at $\alpha=0.05$). The Protein kinase domain appears to be required for retention. A breakpoint-position permutation test produced a corroborating empirical p-value of 0.000999001.

Domain disruption

Domain disruption was tested with Fisher's exact test comparing in-frame fusions against all others; 31/33 (93.9%) of in-frame fusions disrupted the domain, $p=0.0049969$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.0249975.

Cutpoint detection

Cutpoint detection scanned 40 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 439 aa (exon 11; permutation-corrected $p=0.0599401$, not statistically significant at $\alpha=0.05$). This is -19 aa from the nearest configured domain boundary at 458 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=33$) and "not_retained" ($n=7$). For Frame_status, retained: 31/31 (100.0%, 95% CI 89.0%-100.0%); not_retained: 2/2 (100.0%, 95% CI 34.2%-100.0%). Welch's t-test comparing Tumor_variant_count between groups gave means of 48.303 vs 30.8571, $p=0.273788$ (not statistically significant at $\alpha=0.05$).

Composite evidence score

Composite evidence score produced results for this run (25 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Joint partner

Joint partner reported: BRAF has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
SND1	8	0.1951219512195122	0.30004340774793187	0.1602103418604825	0.759046052631579	0.6160624160231085	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.34105265517198086	1
SCRIB	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	1.0	0.6160624160231085	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.32597623508212203	2
PJA2	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.993421052631579	0.6160624160231085	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3249893929768589	3
RBM33	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.993421052631579	0.6160624160231085	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3249893929768589	4
KIAA1549	4	0.0975609756097561	0.30004340774793187	0.1602103418604825	0.7105263157894737	0.6160624160231085	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3045064019627382	5
CDK5RAP2	2	0.04878048780487805	0.30004340774793187	0.1602103418604825	0.805921052631579	0.6160624160231085	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30418146614759056	6
GIPC2	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.6973684210526316	0.6160624160231085	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.28058149824001677	7
KDM7A	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.6973684210526316	0.6160624160231085	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.28058149824001677	8
OSBPL9	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.6973684210526316	0.6160624160231085	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.28058149824001677	9

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
PRKAR2B	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.6973684210526316	0.6160624160231085	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.28058149824001677	10
ZNF207	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.6973684210526316	0.6160624160231085	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.28058149824001677	11
AGAP3	2	0.04878048780487805	0.30004340774793187	0.1602103418604825	0.6151315789473684	0.6160624160231085	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.275563045094959	12
CCDC6	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.618421052631579	0.6160624160231085	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2687393929768589	13
FAM131B	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.618421052631579	0.6160624160231085	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2687393929768589	14
PHTF2	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.618421052631579	0.6160624160231085	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2687393929768589	15
CUL1	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.611842105263158	0.6160624160231085	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2677525508715957	16
LINC00244	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.611842105263158	0.6160624160231085	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2677525508715957	17
METTL2B	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.611842105263158	0.6160624160231085	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2677525508715957	18
PRKAR1B	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.611842105263158	0.6160624160231085	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2677525508715957	19

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
AGK	3	0.07317073170731707	0.30004340774793187	0.1602103418604825	0.26315789473684215	0.6160624160231085	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.23008406563411174	20
MKRN1	3	0.07317073170731707	0.30004340774793187	0.1602103418604825	0.19956140350877194	0.6160624160231085	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.22054459194990123	21
MRPS33	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.26315789473684215	0.6160624160231085	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.21544991929264834	22
PARP12	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	None	0.6160624160231085	['confidence_certainty', 'domain_disruption', 'domain_retention', 'recurrence']	0.2070308648024965	23
VIPR2	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.17105263157894735	0.6160624160231085	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.20163412981896411	24
LUC7L2	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.0	0.6160624160231085	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.175976235082122	25

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
287	0	33	1	6	0.0	0.17500000000000004	0.7569619513137055
327	3	30	2	5	0.25	0.20399751978699343	0.6903751127180751
380	11	22	3	4	0.6666666666666666	0.6789250550860457	0.168178163803398
381	20	13	4	3	1.1538461538461537	1.0	-0.0
393	25	8	4	3	2.34375	0.36939377457953293	0.43251042803096634
438	30	3	5	2	4.0	0.20399751978699346	0.690375112718075
439	33	0	5	2	None	0.026923076923076925	1.569875307956561
565	33	0	6	1	None	0.175	0.7569619513137056

domain_disruption tables

frame_domain_contingency_table

31	3
2	4

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
ras_binding	RAS-binding domain	40	6	0	34	0.0
cysteine_rich	Cysteine-rich domain	40	6	0	34	0.0

domain_retention tables

frame_domain_contingency_table

31	2
2	5

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	40	33	1	6	0.06050420168067227

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
11	34	41	0.8292682926829268

frequency tables

Partner_gene_counts

Partner_gene	Event_count
AGAP3	2
AGK	3
CCDC6	1
CDK5RAP2	2

Partner_gene	Event_count
CUL1	1
FAM131B	1
GIPC2	1
KDM7A	1
KIAA1549	4
LINC00244	1
LUC7L2	1
METTL2B	1
MKRN1	3
MRPS33	1
OSBPL9	1
PARP12	1
PHTF2	1
PJA2	1
PRKAR1B	1
PRKAR2B	1
RBM33	1
SCRIB	1
SND1	8
VIPR2	1
ZNF207	1

Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_prime	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0009757-T01-IM5-1	P-0009757-T01-IM5		BRAF-AGAP3	AGAP3	in-frame	threep_rime	140493871	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) rearrangement: c.1222-2154:AGAP3_c.1140+237:BRAF Protein fusion: in frame (AGAP3-BRAF)	in frame

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0007861-T01-IM5-2	P-0007861-T01-IM5		BRAF-AGAP3	AGAP3	in-frame	three_prime	140490697	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) rearrangement: c.1326+220:AGAP3_c.1141-3313:BRAFInv Protein fusion: in frame (AGAP3-BRAF)	in frame
EVT-P-000913-T02-IM5-3	P-0000913-T02-IM5		BRAF-AGK	AGK	in-frame	three_prime	140496386	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) Fusion (AGK exon 2 fused with BRAF exon 8) : c.101+13932:AGK_c.981-2119:BRAFInv Protein fusion: in frame (AGK-BRAF)	in frame
EVT-P-0010945-T01-IM5-4	P-0010945-T01-IM5		BRAF-AGK	AGK	in-frame	three_prime	140497610	7	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) rearrangement : c.101+12646:AGK_c.980+2552:BRAFInv Protein fusion: in frame (AGK-BRAF)	in frame
EVT-P-0004477-T01-IM5-5	P-0004477-T01-IM5		BRAF-AGK	AGK	in-frame	three_prime	140495405	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exons 1 to 2 and BRAF exons 8 to 18): c.101+10076:AGK_c.981-1138:BRAFInv Protein fusion: in frame (AGK-BRAF)	in frame
EVT-P-0004795-T01-IM5-13	P-0004795-T01-IM5		BRAF-CUL1	CUL1	in-frame	three_prime	140493811	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CUL1(NM_003592) - BRAF (NM_004333) Rearrangement :c.790-427:CUL1_c.1140+297:BRAFInv Protein fusion: in frame (CUL1-BRAF)	in frame

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0001988-T01-I M3-14	P-0001988-T01-IM3		BRAF-FAM131B	FAM131B	in-frame	three_prime	140489629	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		FAM131B (NM_001031690) - BRAF (NM_004333) Deletion : c.175-103:FAM131B_c.1141-2245:BRAFdel Protein fusion: in frame (FAM131B-BRAF)	in frame
EVT-P-0010494-T01-I M5-15	P-0010494-T01-IM5		BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655) -BRAF (NM_004333) Rearrangement : t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-0004871-T01-I M5-16	P-0004871-T01-IM5		BRAF-LINC00244	LINC00244	unknown	five_prime	140492384	8	380	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) rearrangement : c.1140+1724:BRAF_chr7:g.156140327inv BRAF-NCRNA00244 Fusion	NA
EVT-P-0005553-T02-I M5-17	P-0005553-T02-IM5		BRAF-MRPS33	MRPS33	unknown	five_prime	140497546	7	327	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - MRPS33 (NM_053035)Rearrangement : c.980+2616:BRAF_c.215+390:MRPS33inv BRAF-MRPS33 Fusion	NA
EVT-P-0010398-T01-I M5-18	P-0010398-T01-IM5		BRAF-OSBPL9	OSBPL9	in-frame	three_prime	140484912	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		OSBPL9 (NM_148909) - BRAF (NM_004333) Rearrangement : t(1;7)(p32.3;q34)(chr1:g.52228623::chr7:g.140484912) Protein fusion: in frame (OSBPL9-BRAF)	in frame
EVT-P-0003502-T01-I M5-19	P-0003502-T01-IM5		BRAF-PJA2	PJA2	in-frame	five_prime	140482379	10	438	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		NA Protein fusion: in frame (BRAF-PJA2)	in frame

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0009246-T01-IM5-20	P-0009246-T01-IM5		BRAF-RBM33	RBM33	in-frame	three_prime	140482661	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		RBM33 (NM_053043) - BRAF (NM_004333) rearrangement: c.171+526:RBM33_c.1314+160:BRAFinv Protein fusion: in frame (RBM33-BRAF)	in frame
EVT-P-0002516-T01-IM3-21	P-0002516-T01-IM3		BRAF-VIPR2	VIPR2	unknown	five_prime	140454067	14	565	True	disrupted	0.4235294117647059	True	RAS-binding domain; Cysteine-rich domain		Protein kinase domain	NA BRAF-VIPR2 Fusion	NA
EVT-P-0005834-T01-IM5-22	P-0005834-T01-IM5		CCDC6-BRAF	CCDC6	in-frame	three_prime	140488401	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CCDC6 (NM_005436) -BRAFF (NM_004333) Rearrangement : t(10;7)(chr10q21.2;q34)(chr10:g.61627617::chr7:g.140488401) Protein fusion: in frame (CCDC6-BRAF)	in frame
EVT-P-0004203-T01-IM5-23	P-0004203-T01-IM5		CDK5RAP2-BRAF	CDK5RAP2	in-frame	three_prime	140491585	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CDK5RAP2 (NM_018249) - BRAF (NM_004333) rearrangement: t(7;9)(7q34;q33.2)(chr7:140491585::chr9:123251876) Protein fusion: in frame (CDK5RAP2-BRAF)	in frame
EVT-P-0005461-T02-IM5-24	P-0005461-T02-IM5		CDK5RAP2-BRAF	CDK5RAP2	in-frame	three_prime	140481684	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CDK5RAP2 (NM_018249) - BRAF (NM_004333) rearrangement : t(7;9)(7q34;q33.2)(chr7:140481684::chr9:123255452) Protein fusion: in frame (CDK5RAP2-BRAF)	in frame

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001 1111-T01-I M5-25	P-0011 111-T0 1-IM5		KDM7 A-BRAF	KDM7A	unknown	three_prime	140484788	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		BRAF (NM_004333) - JHDM1D (NM_030647) rearrangement: c.1178-1831: BRAF_c.*1672: JHDM1Ddel JHDM1D-BRAF Fusion	NA
EVT-P-000 5121-T02-I M5-26	P-0005 121-T0 2-IM5		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491248	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 with BRAF exons 9-18): c.4929+2454: KIAA1549_c.1140+2860: BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	in frame
EVT-P-000 4032-T01-I M5-27	P-0004 032-T0 1-IM5		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140487892	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exon 16 fused to BRAF exon 9): c.5248-1949: KIAA1549_c.1141-508: BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	in frame
EVT-P-000 9097-T01-I M5-28	P-0009 097-T0 1-IM5		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140488817	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused in frame with BRAF exons 9-18, which include the BRAF kinase domain): c.5247+3837: KIAA1549_c.1141-1433: BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	in frame
EVT-P-000 3664-T01-I M5-29	P-0003 664-T0 1-IM5		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140482817	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		c.4929+549: KIAA1549_c.1314+4: BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	in frame

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0006412-T01-I M5-30	P-0006412-T01-IM5		LUC7L2-BRAF	LUC7L2	unknown	five_prime	140501109	6	287	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		LUC7L2 (NM_001244584) - BRAF (NM_004333) Rearrangement : c.1008-2090:LUC7L2_c.860+103:BRAFdel Antisense fusion	NA
EVT-P-0002305-T01-I M3-31	P-0002305-T01-IM3		METTL2B-BRAF	METTL2B	unknown	three_prime	140491674	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA METTL2B-BRAF Fusion	NA
EVT-P-0004373-T02-I M5-32	P-0004373-T02-IM5		MKRN1-BRAF	MKRN1	unknown	three_prime	140434466	18	744	False	lost	0.0	False		Protein kinase domain; RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.314+4351:MKRN1_c.2232:BRAFdup Protein fusion: mid-exon (MKRN1-BRAF)	NA
EVT-P-0001036-T01-I M3-33	P-0001036-T01-IM3		MKRN1-BRAF	MKRN1	in-frame	three_prime	140482250	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA Protein fusion: in frame (MKRN1-BRAF)	in frame
EVT-P-0009024-T01-I M5-34	P-0009024-T01-IM5		MKRN1-BRAF	MKRN1	in-frame	three_prime	140493485	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.772-184G:MKRN1_c.1140+623:BRAFdup Protein fusion: in frame (MKRN1-BRAF)	in frame

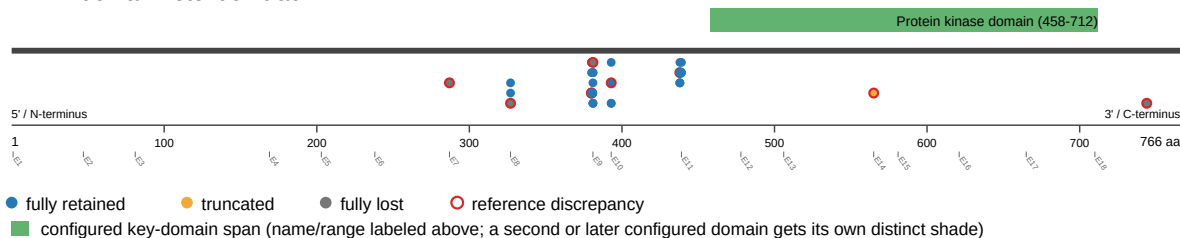
event_id	sampl e_id	pa tie nt _i d	fusion _nam e	part ner _ge ne	fra me _sta tus	tar get _ro le	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_ etained_fr action	domain _is_trun cated	retain ed_do mains	lost_ do mai ns	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-000 8789-T01-I M5-36	P-0008 789-T0 1-IM5		PHTF 2-BRA F	PHT F2	in-fr ame	thr ee_ _pr im e	140489 236	9	381	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		PHTF2 (NM_001127358) - BRAF (NM_004333) rearrangement: c.2224-367: PHTF2_c.1141-1852:BRAFi nv Protein fusion: in frame (PHTF2-BRAF)	in frame
EVT-P-000 9694-T01-I M5-37	P-0009 694-T0 1-IM5		PRKA R1B-B RAF	PRK AR1 B	in-fr ame	thr ee_ _pr im e	140491 855	8	380	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		PRKAR1B (NM_001164758) -BRAF (NM_004333) Rearrangement : c.892-255 1:PRKAR1B_c.1140+2253: BRAFDup Protein fusion: in frame (PRKAR1B-BRAF)	in frame
EVT-P-000 8567-T01-I M5-38	P-0008 567-T0 1-IM5		PRKA R2B-B RAF	PRK AR2 B	in-fr ame	thr ee_ _pr im e	140486 416	9	393	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		PRKAR2B (NM_002736) - BRAF (NM_004333) rearrangement: c.307+8063: PRKAR2B_c.1177+932:BR AFinv Protein fusion: in frame (PRKAR2B-BRAF)	in frame
EVT-P-000 7392-T01-I M5-39	P-0007 392-T0 1-IM5		SCRIB-BR AF	SCR IB	in-fr ame	thr ee_ _pr im e	140482 150	11	439	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		SCRIB (NM_182706) - BRAF (NM_004333) rearrangement: t(7:8)(q34;q 24.3)(chr7:g.140482150::chr 8:g.144876675) Protein fusion: in frame (SCRIB-BRAF)	in frame
EVT-P-000 4103-T01-I M5-40	P-0004 103-T0 1-IM5		SND1- BRAF	SND 1	in-fr ame	thr ee_ _pr im e	140481 685	11	439	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1038+579 9:SND1_c.1315-192:BRAFi nv. Protein fusion: in frame (SND1-BRAF)	in frame

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000 9619-T01-I M5-41	P-0009 619-T0 1-IM5		SND1- BRAF	SND 1	in-frame	three_prime	140482 494	10	438	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1968+2394:SND1_c.1314+327:BRAFInv Protein fusion: in frame (SND1-BRAF)	in frame
EVT-P-001 2395-T01-I M5-42	P-0012 395-T0 1-IM5		SND1- BRAF	SND 1	in-frame	three_prime	140493 527	8	380	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1152+27174:SND1_c.1140+581:BRAFInv Protein fusion: in frame (SND1-BRAF)	in frame
EVT-P-000 8140-T01-I M5-43	P-0008 140-T0 1-IM5		SND1- BRAF	SND 1	in-frame	three_prime	140489 439	9	381	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-9 fused with BRAF exons 9-18) : c.1039-3849:SND1_c.1141-2055:BRAFInv Protein fusion: in frame (SND1-BRAF)	in frame
EVT-P-000 2197-T01-I M3-44	P-0002 197-T0 1-IM3		SND1- BRAF	SND 1	in-frame	five_prime	140488 303	9	381	True	lost	0.0	False	RAS-binding domain; Cysteine- rich domain	Protein kinase domain		c.1528-8436:SND1_c.1141-922:BRAFInv Protein fusion: in frame (BRAF-SND1)	in frame
EVT-P-000 2071-T02-I M5-45	P-0002 071-T0 2-IM5		SND1- BRAF	SND 1	in-frame	three_prime	140489 292	9	381	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 fused with BRAF exons 9-18) : c.1152+18606:SND1_c.1141-1908:BRAFInv Protein fusion: in frame (SND1-BRAF)	in frame

event_id	sampl e_id	pa tie nt _i d	fusion _nam e	part ner _ge ne	fra me _sta tus	tar get _ro le	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_ etained_fr action	domain _is_trun cated	retain ed_do mains	lost_ do mai ns	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-000 7533-T01-I M5-46	P-0007 533-T0 1-IM5		SND1- BRAF	SND 1	in-fr ame	thr ee_ pr ime	140482 226	10	438	True	retain ed	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 with BRAF exons 11-18) : c.1152+31738:SND1_c.1314+595:BRANV Protein fusion: in frame (SND1-BRAF)	in frame
EVT-P-000 1687-T01-I M3-47	P-0001 687-T0 1-IM3		SND1- BRAF	SND 1	in-fr ame	thr ee_ pr ime	140490 224	9	381	True	retain ed	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		SND1 (NM_014390) - BRAF (NM_004333) inversion: c.1152+28381_c.1141-2840inv Protein fusion: in frame (SND1-BRAF)	in frame
EVT-P-000 8920-T01-I M5-48	P-0008 920-T0 1-IM5		ZNF2 07-BR AF	ZNF 207	in-fr ame	thr ee_ pr ime	140485 847	9	393	True	retain ed	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		ZNF207 (NM_001098507) - BRAF (NM_004333) rearrangement: t(7;17)(q34;q11.2)(chr7:g.140485847::chr17:g.30687533) Protein fusion: in frame (ZNF207-BRAF)	in frame

domain retention outliers

BRAF domain-retention track



fusion schematic

BRAF fusion-transcript schematic

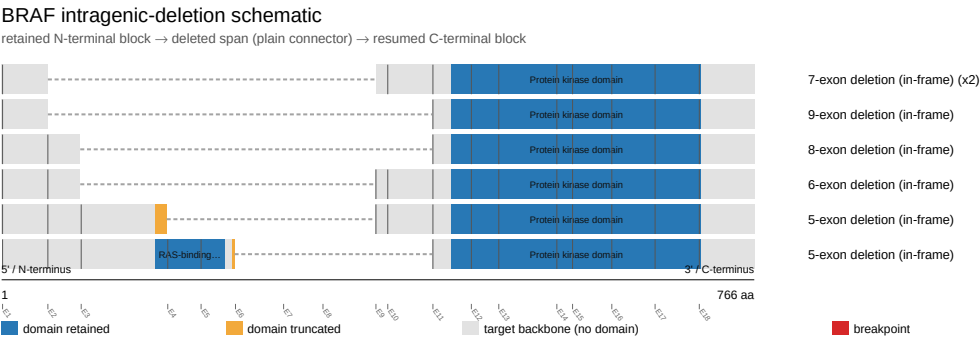
Fusion order varies by row; follow each row's 5' and 3' labels (BRAF portion is domain-colored, exon ticks)



Partner-block colors are decorative; row labels identify partners.

Showing the top 28 of 34 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

intragenic deletion schematic



reference comparison

Reference vs msk_impact_2017

